# FOREST WEATHER INDEX (FWI) PREDICTION SYSTEM USING MACHINE LEARNING

## SECTION 1 : INTRODUCTION


### ABSTRACT

Forest fires are one of the most critical environmental hazards, causing severe damage to forests, wildlife, and human settlements. Accurate prediction of fire-prone conditions can significantly reduce losses by enabling early preventive measures. This project focuses on predicting the Forest Weather Index (FWI) using machine learning techniques. Historical meteorological and fire index data are analyzed and preprocessed before training a Ridge Regression model to estimate FWI values. The trained model is integrated into a Flask-based web application that allows users to input weather parameters and obtain real-time fire risk predictions through an interactive and visually enhanced interface.


### PROBLEM STATEMENT

Traditional forest fire risk assessment methods often rely on manual observation and static rule-based systems, which may fail to capture complex relationships among environmental factors. These approaches are time-consuming, less scalable, and prone to inaccuracies under changing climatic conditions. There is a need for an automated, data-driven system that can analyze historical fire data and accurately predict forest fire risk using modern machine learning techniques.

### OBJECTIVES

The primary objectives of this project are:

- To analyze a real-world forest fire dataset and identify key factors influencing fire risk.
- To preprocess the dataset using appropriate standardization and normalization techniques.
- To build and tune a Ridge Regression model for predicting the Forest Weather Index.
- To perform exploratory data analysis using statistical measures and visualizations.

- To deploy the trained model using a Flask web application.
- To provide an intuitive and modern user interface for real-time fire risk prediction.

## SECTION 2: DATASET DESCRIPTION AND PREPROCESSING

### DATASET DESCRIPTION

The project uses the Algerian Forest Fires Dataset, which contains meteorological observations and fire weather indices collected from forest regions in Algeria. The dataset includes daily records related to weather conditions and fuel moisture indices, which are widely used in forest fire risk assessment systems. The target variable in this dataset is the Forest Weather Index (FWI), which represents the overall fire danger level.

### FEATURE DESCRIPTION

The dataset consists of the following features:

- **Temperature (°C):** Measures the ambient air temperature.
- **Relative Humidity (RH %):** Indicates the amount of moisture present in the air.
- **Wind Speed (Ws km/h):** Represents wind intensity, which affects fire spread.
- **Rain (mm):** Records the amount of rainfall.
- **FFMC (Fine Fuel Moisture Code):** Indicates moisture content of surface litter.
- **DMC (Duff Moisture Code):** Represents moisture content of moderately deep organic layers.
- **DC (Drought Code):** Reflects long-term dryness of deep soil layers.
- **ISI (Initial Spread Index):** Measures the expected rate of fire spread.
- **BUI (Buildup Index):** Represents the amount of fuel available for combustion.
- **FWI:** Target variable representing overall fire danger.

## PREPROCESSING TECHNIQUES

The Forest Weather Index dataset contains multiple features measured in different units and scales. For example, temperature is measured in degrees Celsius, rainfall is measured in millimeters, wind speed is measured in kilometers per hour, and fire indices such as FFMC, DMC, and DC have entirely different numerical ranges. If such data is directly used for model training, features with larger numerical values may dominate the learning process, leading to biased predictions.

To address this issue and ensure fair contribution from all features, standardization and normalization techniques were applied during preprocessing.

## STANDARDIZATION

Standardization is applied to rescale the features such that they have a mean of zero and a standard deviation of one. This step is essential because the dataset contains features with different scales. Standardization ensures that no single feature dominates the learning process and improves the stability and convergence of the Ridge Regression model.

In this project, standardization helped in the following ways:

- It ensured that meteorological features and fire indices with different magnitudes were treated equally.
- It improved the numerical stability of the Ridge Regression model.
- It enhanced convergence during model training by preventing features with larger values from dominating the loss function.
- It made the regularization process in Ridge Regression more effective by allowing the penalty term to act uniformly across all coefficients.

As Ridge Regression relies on minimizing a cost function with a regularization term, standardized inputs are essential for meaningful coefficient optimization.

## NORMALIZATION

Normalization is used to rescale feature values into a fixed range. This helps in maintaining numerical stability during training and ensures consistent feature representation. Together with standardization, normalization improves model performance and reliability.

In this project, normalization contributed by:

- Reducing the effect of outliers on model performance.
- Improving the stability of predictions across different input ranges.
- Ensuring consistent feature scaling when user inputs are provided through the Flask web application.
- Making the deployed model robust to real-time input variations.

Normalization also supports seamless integration between the preprocessing pipeline and the web application, ensuring that user-provided inputs are scaled in the same manner as the training data.

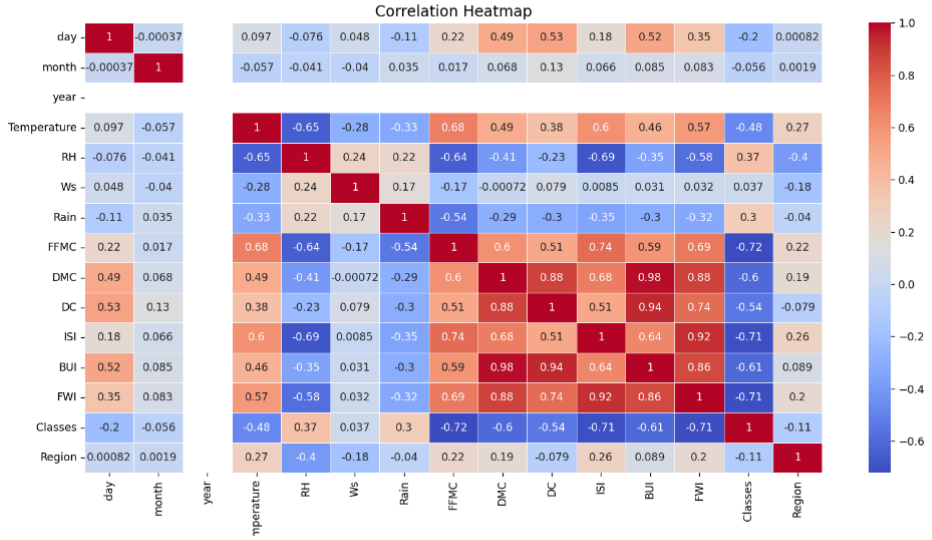
## SECTION 3: EXPLORATORY DATA ANALYSIS (EDA)

### DATASET STATISTICS

Descriptive statistical measures such as mean, median, standard deviation, minimum, and maximum values were computed for all features. These statistics provide insights into the central tendency, spread, and variability of the dataset. Understanding these values helps in identifying outliers and selecting appropriate preprocessing techniques.

### CORRELATION ANALYSIS

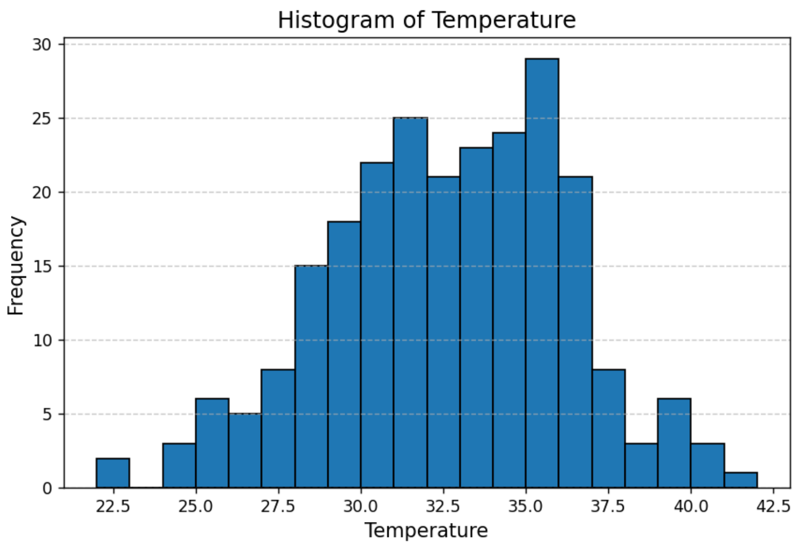
A correlation heatmap was generated to examine the relationships between different features and the target variable. The analysis revealed strong correlations between certain fire weather indices such as FFMC, ISI, BUI, and the Forest Weather Index. These correlations indicate that fuel moisture and spread-related indices play a significant role in fire risk prediction.




- Strong inter-feature correlations indicate the presence of multicollinearity in the dataset, which justifies the use of Ridge Regression to improve model stability through regularization.

**HISTOGRAM ANALYSIS**

Histograms were plotted for each feature to visualize their data distributions. The histograms helped identify skewness, concentration of values, and the presence of outliers. Some features showed right-skewed distributions, highlighting the need for scaling and normalization before model training.



- The temperature values are mostly concentrated between moderate to high ranges, indicating that forest fire occurrences are more frequent under warm weather conditions.
- The spread and slight skewness in the temperature distribution highlight the need for feature scaling to ensure consistent model learning and accurate prediction.

## SECTION 4 : MODEL DEVELOPMENT

### RIDGE REGRESSION

Ridge Regression is a linear regression technique that includes L2 regularization. The regularization term penalizes large coefficient values, reducing model complexity and preventing overfitting. Ridge Regression is particularly effective when multicollinearity exists among features, as observed in this dataset.

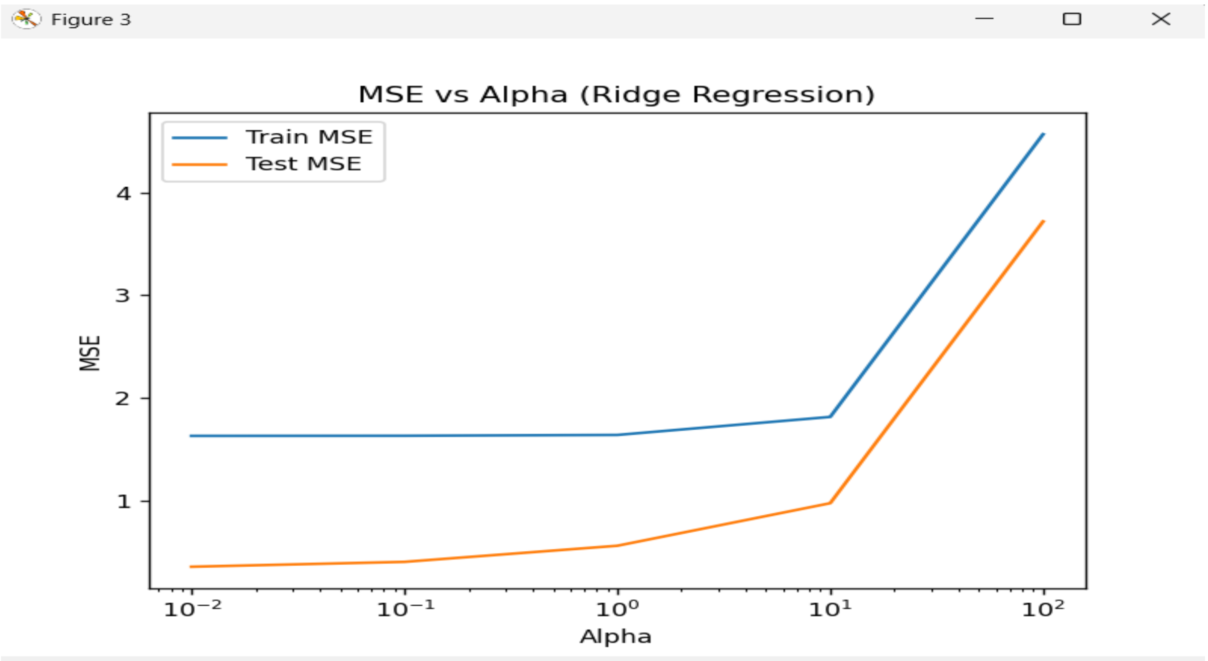
### TUNABLE PARAMETER : ALPHA

The alpha parameter controls the strength of regularization in Ridge Regression. A smaller alpha value results in a model similar to ordinary linear regression, while a larger alpha increases regularization strength. Multiple alpha values were tested to identify the optimal balance between bias and variance.

### MODEL PERFORMANCE EVALUATION

A plot of Mean Squared Error (MSE) versus alpha was generated to analyze model performance across different regularization strengths. The plot showed that the error decreased initially and then increased after a certain alpha value, indicating the optimal alpha where the model achieved minimum error.

Mean Absolute Error (MAE) was also computed to measure the average absolute difference between predicted and actual FWI values. MAE provided an intuitive measure of prediction accuracy and confirmed the effectiveness of the chosen model.




**SECTION 5: WEB APPLICATION DEVELOPMENT**

**FLASK FRAMEWORK**

Flask is a lightweight Python web framework used to deploy the trained machine learning model. It provides flexibility and simplicity for building web-based ML applications.

**BACKEND INTEGRATION WITH FLASK**

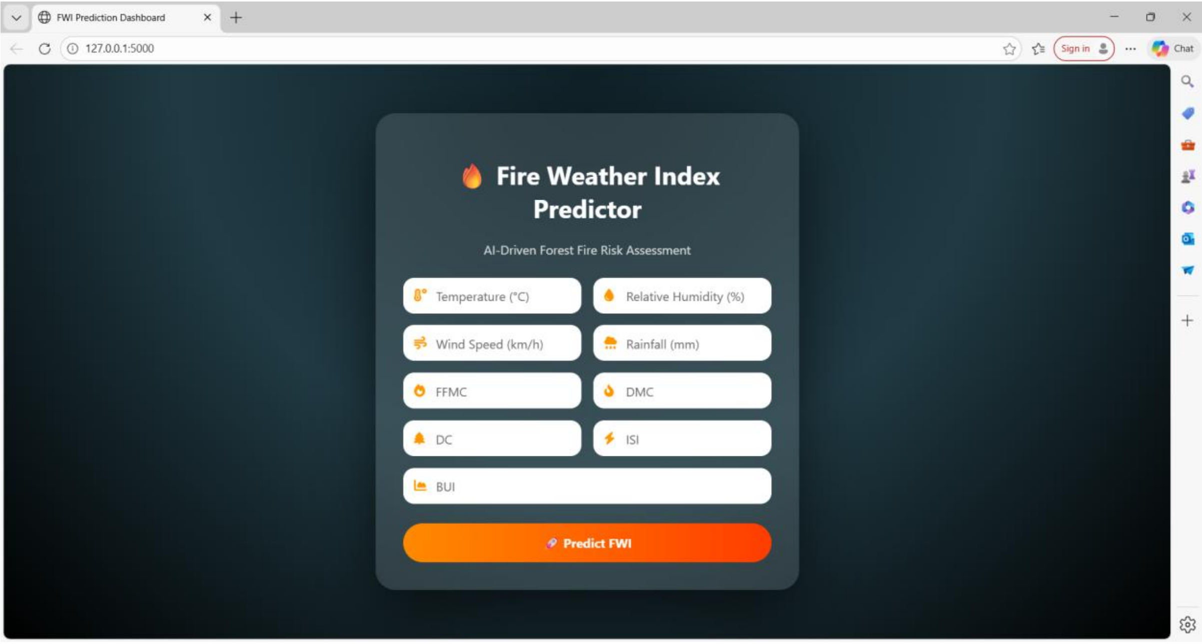
The trained Ridge Regression model and preprocessing scaler were saved using the joblib library. In the Flask backend, these artifacts are loaded at runtime. User input values from the web interface are processed, standardized, and passed to the model for prediction. The predicted FWI value is then classified into different risk levels and returned to the frontend.

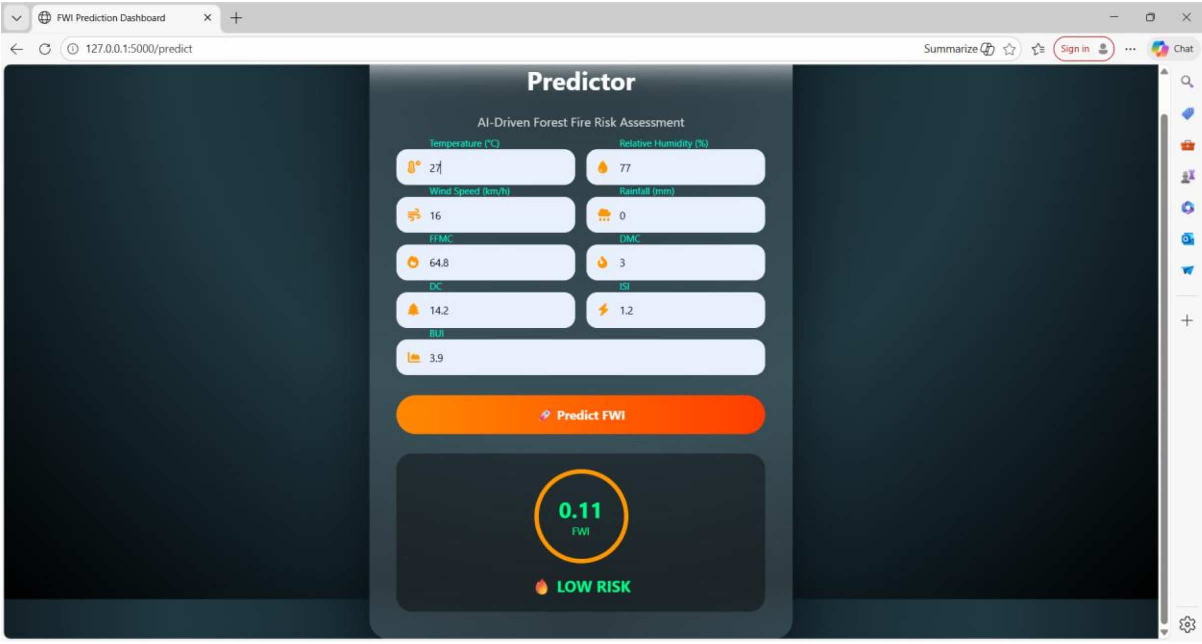
**FRONTEND AND BACKEND INTERACTION**

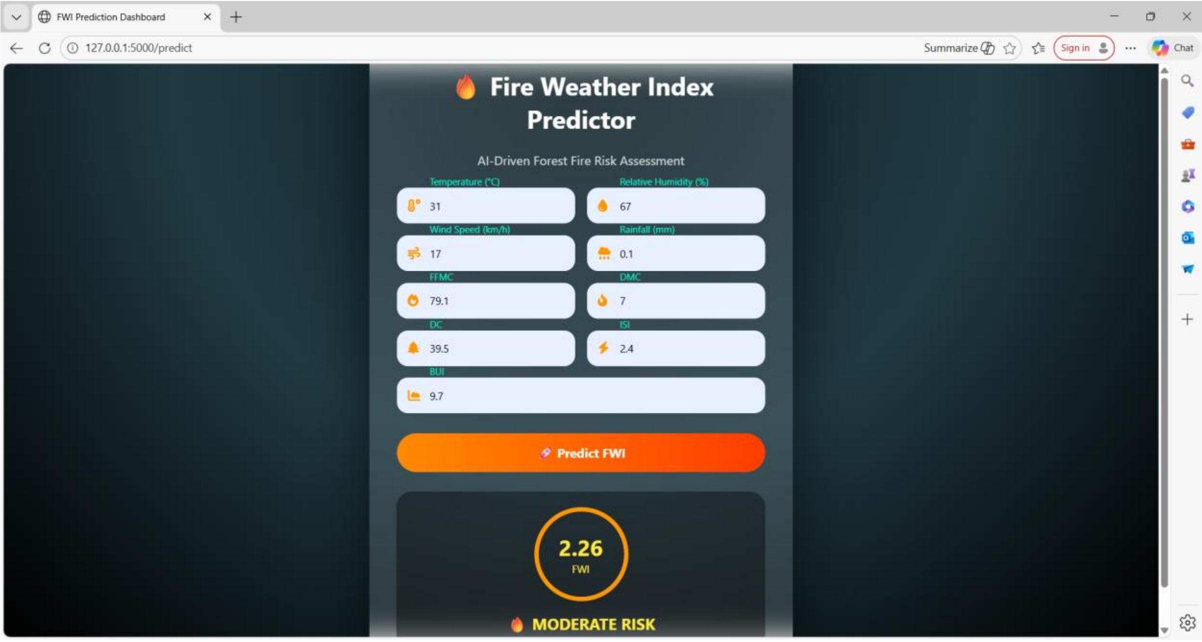
The frontend collects user inputs through a structured form and sends them to the Flask backend via HTTP POST requests. The backend processes the inputs and sends the prediction results back to the frontend, where they are displayed dynamically along with risk indicators and visual cues.

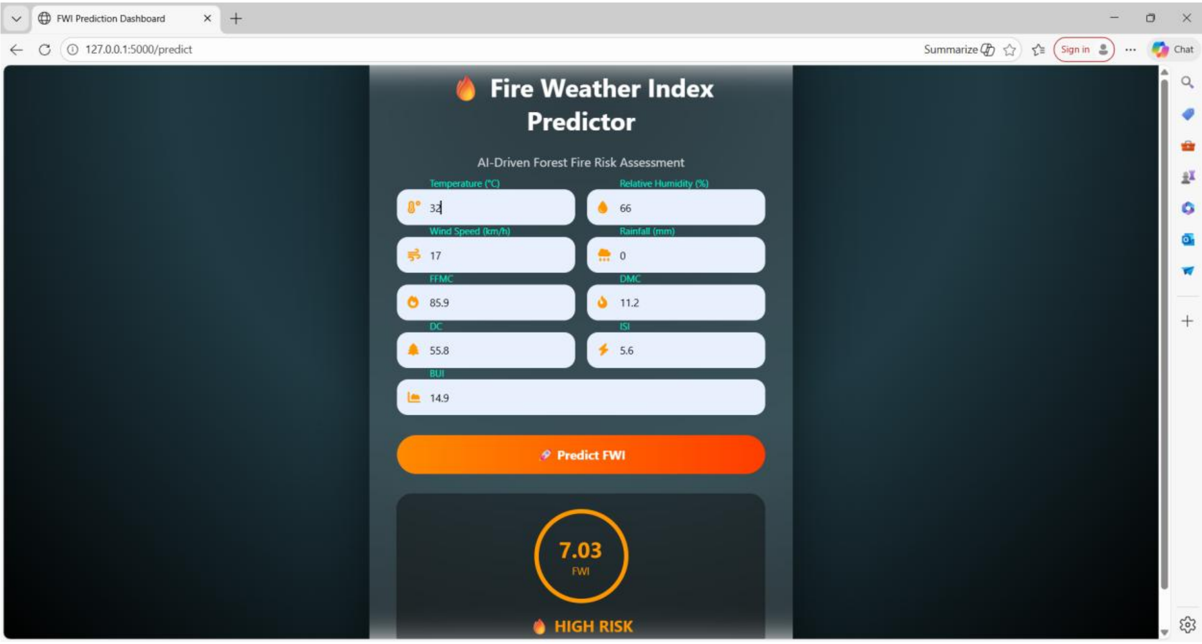
## SECTION 6: DEMO AND OUTPUTS

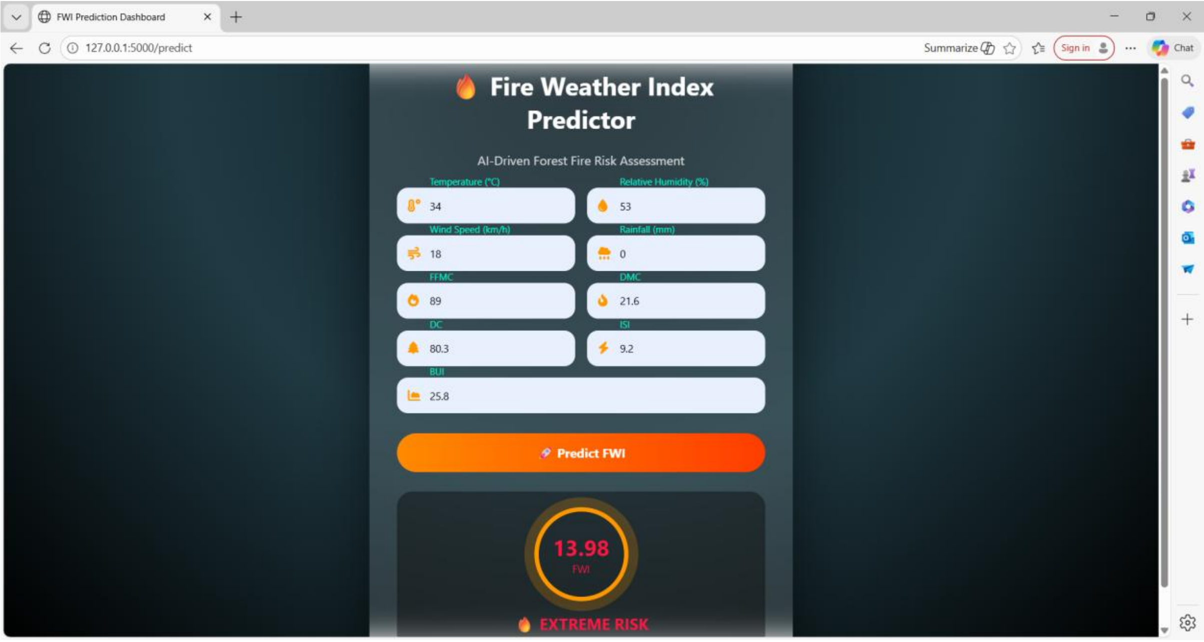
This section demonstrates the working of the Forest Weather Index Prediction System. Screenshots of the web application interface, input forms, and prediction results are included to showcase real-time prediction and risk classification.




FWI Prediction Dashboard
127.0.0.1:5000/predict
Summarize
Sign in
Chat
Fire Weather Index Predictor
AI-Driven Forest Fire Risk Assessment
Temperature (°C)
31
Relative Humidity (%)
67
Wind Speed (km/h)
17
Rainfall (mm)
0.1
FFMC
79.1
DMC
7
DC
39.5
ISI
2.4
BUI
9.7
Predict FWI
2.26
FWI
MODERATE RISK


FWI Prediction Dashboard
127.0.0.1:5000/predict
Summarize
Sign in
Chat
Fire Weather Index Predictor
AI-Driven Forest Fire Risk Assessment
Temperature (°C)
32
Relative Humidity (%)
66
Wind Speed (km/h)
17
Rainfall (mm)
0
FFMC
85.9
DMC
11.2
DC
55.8
ISI
5.6
BUI
14.9
Predict FWI
7.03
FWI
HIGH RISK




## RESULTS AND ANALYSIS

The Ridge Regression model demonstrated reliable performance in predicting Forest Weather Index values. Exploratory data analysis and feature correlations supported the selection of relevant input parameters. The integration of machine learning with a Flask web application provided a practical and user-friendly solution for fire risk prediction.

## CONCLUSION

This project successfully implements a machine learning-based Forest Weather Index prediction system. By combining data preprocessing, exploratory analysis, Ridge Regression modeling, and web deployment, the system provides an efficient approach to forest fire risk assessment. The project highlights the practical application of machine learning in environmental monitoring and can be extended further using advanced models and real-time data sources.